

1. Dataset Staging

AlphaDent
(Kaggle mount)
Zenodo mirror fallback

Recursive copy
to writable dir:
images, labels, YAML

2. YOLO26 Training

Dynamic GPU set
 $\mathcal{D} = \{0, \dots, N_{\text{GPU}} - 1\}$

DDP training ($\approx 1.85\times$)
26m@640,
26l/26x@960
Segment26 head

3. Memory-Safe Validation

Single device $\mathcal{D}_0$;
flush cache,
no raw dumps

Self-healing fallback:
CUDA OOM $\Rightarrow$ re-run
`val()` on CPU

4. Prediction & Eval

9-class pathology
instance masks

Box / Mask mAP@50
evaluation

5. Explainability (XAI)

YOLOExplainer: C3k2 neck
hooks (idx 22) $\rightarrow A^k$;
Segment26 back-
prop $\partial y^c / \partial A^k$

7 CAM methods:
Eigen, Grad, Grad++,
XGrad, HiRes,
Layer, EigenGrad

Quantitative
eval: faithfulness
(del/ins AUC) + agreement
 $\Rightarrow$ Eigen-CAM most faithful

Clinician visual audit of
lesion attributions